ZS-M12

Auto-Surgery: Singularity Resolution via i-Tetration Dynamics

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Verification: 35/35 PASS | Zero Free Parameters

§0. Abstract

We derive the continuous flow equation $d\Phi/d\tau = i^N - \Phi$ from the Z-sector transfer map $T(z) = i^z$ (ZS-M1 HSI Theorem) and prove that $z^* = 0.4383 + 0.3606i$ is its unique globally attracting damped spiral (half-life 0.44 τ_P, DERIVED). The σ-HSI Locking Theorem identifies z^* as the unique joint conformal/stroboscopic fixed point with $c = \dim(Z) = 2$ and $\Delta = 1.885$ (DERIVED-CONDITIONAL). The topological cap $\Omega^2 = 1 + A \cdot \eta_{topo} = 1.0258$ resolves singularities dynamically (Auto-Surgery), filling NC-A6.3.

A No-Go Theorem proves the minimal homogeneous model cannot launch inflation from z^*. The post-bounce Quantum Foam Engine (Y→X micro-BH →Y Hawking radiation) creates a thermal bath at $T \sim 0.41$ MP with high-temperature symmetry restoration. The critical temperature $T_c \approx 2.48 \times 10^{15}$ GeV coincides with the ZS-U2 reheating temperature (cyclic self-consistency). Z-Telomere phase accumulation $\delta\varphi = A$ generates a conserved U(1) comoving charge $Q = A$. The resulting centrifugal barrier:

$$\varepsilon_{min} = (Q^2/\lambda)^{1/6} \approx 30.7 \gg 2.64 = \varepsilon_{sr}$$

prevents the field from reaching $\varepsilon = 0$ at the bounce. The barrier weakens as a^{-1} (launch-then-release), naturally handing off to ZS-U1 slow-roll. If the comoving charge survives the Planck-to-bulk handoff (V1, OPEN), the inflation initial-condition problem is resolved with zero new parameters. Five hypotheses (H2, H5, H6, H7, H2+H5) were systematically tested and rejected en route to this discovery. Verification: 35/35 PASS.

Keywords: i-tetration, singularity resolution, centrifugal barrier, inflation initial conditions, U(1) angular momentum, quantum foam, Z-Telomere, cyclic cosmology, Z-Spin

Epistemic Status Legend

Status	Definition

PROVEN	Follows from standard mathematics alone.
DERIVED	Follows from Z-Spin action + standard physics. Zero free parameters.
DERIVED-CONDITIONAL	Derived, conditional on a stated assumption.
DISPROVEN	Tested and quantitatively rejected.
HYPOTHESIS	Motivated conjecture; derivation incomplete.
NON-CLAIM	Explicitly not asserted.
OPEN	Identified problem requiring investigation.

§1. Introduction

This paper establishes Auto-Surgery—a dynamical singularity-resolution mechanism within Z-Spin Cosmology governed by the single geometric impedance **A = 35/437**—and traces its consequences to the inflation initial condition problem.

The research trajectory is worth noting. The initial Perelman–Ricci flow analogy motivated the continuous flow derivation (§2) and σ-HSI Locking Theorem (§3). The No-Go Theorem (§5) then proved the minimal model cannot launch inflation. Five candidate mechanisms—parametric resonance (H2), Y-sector amplification (H5), spiral-to-radius transduction (H6), Planck thermalization (H7), and H2+H5 combined—were systematically tested and rejected (§8). The Quantum Foam Engine hypothesis (§6) revealed high-temperature symmetry restoration and the Tc ≈ Treh coincidence. Analysis of the domain wall problem uncovered the centrifugal barrier (§7): U(1) angular momentum from Z-Telomere phase accumulation prevents the field from reaching ε = 0, resolving the inflation initial condition problem contingent on a single matching question (V1).

§1.1 Locked Inputs

Quantity	Value	Source	Status
A	35/437	ZS-F2	LOCKED
z*	0.4383+0.3606i	ZS-M1	PROVEN
η_topo = $\lvert z^*\rvert^2$	0.3221	ZS-M1	PROVEN
$\lvert f'(z^*)\rvert$	0.8915	ZS-M1	PROVEN
δφ = A	Phase per cycle	ZS-U5	DERIVED-under-P6
y_eff	35/472	ZS-U2	DERIVED
m_eff	3.69×10^{-6} M_P	ZS-U1	DERIVED
λ	7.63×10^{-12}	ZS-U1	EXTERNAL

§2. Continuous i-Tetration Flow

From $T(z) = i^z$ (HSI Theorem, ZS-M1) via forward-Euler correspondence with step $h = \tau P$:

$$\mathbf{d\Phi/d\tau = i^N - \Phi \quad (1)}$$

Theorem 2.1. z^* is the unique attractive fixed point: damped spiral, $\lambda_{1,2} = -1.566 \pm 0.688i$, half-life 0.44 τP, 99% convergence 2.94 τP. [PROVEN]

Correction: The Langevin-type flow $d\Phi/d\tau = \ln(i) \cdot i^N - \Phi$ has a different fixed point $z_2 \approx 0.606i \neq z^*$. Only Eq. (1) is HSI-consistent. [PROVEN]

§3. σ-HSI Locking Theorem

Theorem 3.1. Under H-M12.1 (Z-sector ≡ 2D σ-model), the unique state satisfying (i) $\beta^G = 0$ and (ii) $i^N = \Phi$ is $\Phi = z^*$. $c = \dim(Z) = 2$, $\Delta = 1.885$. [DERIVED-CONDITIONAL]

§4. Auto-Surgery Mechanism

Four stages: (1) Normal KG, (2) $R \to \infty$, (3) i-tetration: $\Phi \to z^*$ in ~3 τP, (4) Topological cap $\Omega^2 = 1.0258$, $G_{eff} = 0.9749G$ (finite, DERIVED). Structural correspondence with Perelman's surgery (HYPOTHESIS).

§5. No-Go: Minimal Model Cannot Launch Inflation

Theorem 5.1 (Lyapunov). $L(\Phi) = |\Phi - z^*|^2$ satisfies $dL/d\tau = 2|\delta|^2 \times (-1.566) < 0$. [PROVEN]

Theorem 5.2 (Curvature Bound). $\Delta\varepsilon \leq 0.21$ during ~3 τP surgery. Max reach $0.78 \ll \varepsilon_{sr} = 2.64$. [DERIVED]

Inverse Problem. Minimum energy $\Delta V = 2.72 \times 10^{-11}\ M^4_P$ (27 parts per trillion of Planck energy). The bottleneck is coherence, not energy. [DERIVED]

§6. Quantum Foam Engine and Thermal Phase Transition

§6.1 Y → X → Y Energy Cycle

Post-bounce energy creates Planck-mass micro-BH (capped by $\Omega^2 = 1.0258$). Hawking evaporation ($T_H \approx 0.04$ MP, $\tau_{evap} \approx 16{,}000\ \tau_P$) returns 99% to Y-sector particles (76% fermions, 23% gauge bosons). Thermal bath at $T \approx 0.41$ MP. [HYPOTHESIS (strong)]

§6.2 High-Temperature Symmetry Restoration

Thermal mass $\Delta m^2_{th} \approx 1.24\times10^{-6}\ M^2_P$ (95,000× bare mass) triggers symmetry restoration: only $\varepsilon = 0$ minimum survives. Critical temperature:

$$\mathbf{T_c = \sqrt{(16\pi^2\lambda/(A^2 b_{eff}))} \approx 2.48\times10^{15}\ GeV \approx T_{reh} \quad (2)}$$

The $T_c \approx T_{reh}$ coincidence is not accidental: both are controlled by A and λ. Tachyonic growth rate $m_{tach}/H = 1.35 > 1$ at T_c. [DERIVED]

§7. Centrifugal Barrier: Inflation Initial Condition

§7.1 U(1) Comoving Charge from Z-Telomere

For $\Phi = \varepsilon e^{i\theta}$ in FRW, the conserved U(1) Noether charge is $Q = a^3\varepsilon^2\dot{\theta}$ (comoving). At the Z-Telomere onset ($a = 1$, $\varepsilon = 1$, $\dot{\theta} = A$ from ZS-U5): $Q = A = 35/437$. The U(1) has no anomaly (perturbative, non-perturbative, or gravitational—scalar, no chiral fermions). [PROVEN]

§7.2 Effective Potential with Centrifugal Term

With $Q \neq 0$: $\dot{\theta} = Q/(a^3\varepsilon^2)$, giving:

$$\mathbf{V_{eff}(\varepsilon, a) = (\lambda/4)(\varepsilon^2-1)^2 + Q^2/(2a^6\varepsilon^2) \quad (3)}$$

Setting $V'_{eff} = 0$ for large ε: $\lambda\varepsilon^3 = Q^2/(a^6\varepsilon^3)$, hence $\varepsilon^6 = Q^2/(\lambda a^6)$, giving:

$$\mathbf{\varepsilon_{min}(a) = (Q^2/(\lambda a^6))^{1/6} = (Q^2/\lambda)^{1/6} / a \quad (4)}$$

At the bounce ($a = 1$): $\varepsilon_{min} = ((35/437)^2 / 7.63\times10^{-12})^{1/6} = (8.41\times10^8)^{1/6} \approx$ **30.7**. Numerical verification: $V'_{eff} = 0$ root at $\varepsilon = 30.73$ (0.02% agreement). [DERIVED-CONDITIONAL]

§7.3 Launch-Then-Release Dynamics

The barrier $\varepsilon_{min}(a) = 30.7/a$ weakens as the universe expands:

ε_min(a)	a	Interpretation
30.7	1 (bounce)	Deep large-field regime
15.4	2	Still large-field
6.1	5	Above slow-roll
3.1	10	Near ε_sr
2.64	11.6 (critical)	Barrier falls below ε_sr

This is physically superior to a permanent barrier: the centrifugal term *launches* the field at large ε, then *releases* as a grows, allowing ordinary ZS-U1 slow-roll to take over. ZS-U1 operates at $\varepsilon_0 \sim 20$ (N ~ 60 e-folds), consistent with εmin(a=1) = 30.7 > 20. [DERIVED-CONDITIONAL]

§7.4 Robustness: L Survival Through Auto-Surgery

The i-tetration flow dissipates angular momentum. The corrected threshold:

$$\mathbf{Q_{threshold} = \sqrt{\lambda} \times \varepsilon^3_{sr} = 5.08\times10^{-5} \quad (5)}$$

Safety margin: Qinitial/Qthreshold = $A/(5.08\times10^{-5})$ = **1,576**. Dissipation tolerance: 99.94%. The i-tetration decay rate $\gamma = 1.566/\tau_P$ (PROVEN, eigenvalue) gives $\tau_{critical} = \ln(1576)/1.566$ = **4.7 τP**. Since the Planck-bulk transition occurs at τ ~ 1–3 τP, the safety margin is 1.6×.

Transition τ	Q residual	ε_min (1/6)	> ε_sr?
1 τ_P (10.7%)	0.0085	14.6	✓ (5.5×)
2 τ_P (~1%)	~0.0008	6.7	✓ (2.5×)
3 τ_P (0.1%)	~0.00008	3.1	✓ (1.2×)
4.7 τ_P (threshold)	5.1×10^{-5}	2.64	= ε_sr

Frame transformation: $Q_E = Q_J \times \sqrt{(1+A\varepsilon^2)} \approx 1.04 \times Q_J$ (~4% correction, frame-independent). [DERIVED]

§7.5 Domain Wall Resolution

At Tc, Npatches = $(R_{bounce}/R_H(T_c))^3 \approx 5.6\times10^{-10} < 1$: single causal region at the phase transition. Combined with the centrifugal barrier (ε never reaches 0), no domain walls form. [DERIVED]

§7.6 The Decisive Open Question: V1

If the post-bounce bulk U(1) comoving charge Q survives the Planck-to-bulk handoff, then the centrifugal launch mechanism naturally places ε in the large-field regime required by ZS-U1. The remaining inflation

initial-condition problem is reduced to a single matching question: does $Q = a^3\varepsilon^2\dot{\theta}$, accumulated during Z-Telomere dynamics, survive the ~3 τ_P dissipative Auto-Surgery phase? [STATUS: V1 OPEN]

§8. Tested and Rejected Hypotheses

Hypothesis	Mechanism	Result	Status
H2	Parametric resonance	Energy flows FROM ε (depletion)	DISPROVEN
H5	Y-sector amplification	Modifies potential, no energy injection	DISPROVEN
H2+H5	Combined	Both insufficient	DISPROVEN
H6	Spiral-to-radius transduction	Dissipative flow: $Re(\lambda)<0$, amplitude non-increasing	DISPROVEN
H7	Planck thermalization	Four critiques: formula, zero-mode, handoff, frame	WEAKENED

The systematic rejection of five hypotheses led to the discovery of the centrifugal barrier mechanism, this paper's principal result.

§9. Complete Cyclic Cosmology

Phase E (Auto-Surgery): Singularity resolved. $\Omega^2 = 1.0258$. ~3 τ_P.
Phase E′ (Quantum Foam): Micro-BH → Hawking → T ~ 0.41 MP. Symmetry restoration.
Phase E″ (Centrifugal Launch): Q = A → $\varepsilon_{min} = 30.7/a$. Field launched to large-field regime.
Phase F (Inflation): ZS-U1 slow-roll takes over as barrier weakens. $n_s = 0.9674$, $r = 0.0089$.

§10. Non-Claims

NC-M12.1 (V1): The centrifugal barrier does NOT claim that Q survives Auto-Surgery. V1 (Planck-bulk matching) is OPEN. The NC-M12.1 inflation IC problem is reduced to this single matching question, not resolved.

NC-M12.2: The σ-HSI Locking Theorem is conditional on H-M12.1 (Z-sector = 2D σ-model).

NC-M12.3: The Quantum Foam Engine is HYPOTHESIS. Planck-scale BH formation is beyond direct simulation.

NC-M12.4: Q = A assumes $\dot{\theta} = A$ at Z-Telomere onset (DERIVED-under-P6). Modification of $\delta\varphi$ changes ε_{min}.

§11. Falsification Gates

Gate	Description	Type
F-M12.1	Second attractive fixed point of Eq. (1)	Mathematical
F-M12.2	Z-sector ≠ 2D σ-model	Theoretical
F-M12.3	Coupled NR: curvature unbounded	Computational
F-M12.4	U(1) broken by quantum gravity	Theoretical
F-M12.5	$\delta\varphi \neq A$ (P6 revision)	Foundational
F-M12.6	CMB-S4: no bounce signatures	Observational

§12. Verification: 35/35 PASS

Category	Tests	Description
A. Constants	3	A, z*, η_topo, |f′(z*)|
B. Flow	4	Fixed point, Jacobian, convergence, basin
C. σ-Model	3	R_target=0, c=2, Δ=1.885
D. Auto-Surgery	5	Cap, G_eff, duration, V(|z*|), hierarchy
E. Cross-Paper	3	ZS-M1, ZS-A6, ZS-U1
F. Complementarity	4	S_flow≠S_tunnel, dim, ratio, index
G. Thermalization	5	m_eff, T_c, V_plateau/ρ_P, H2 q, overshoot
H. Centrifugal	8	ε_min(1/6)=30.7, Q_th, margin=1576×, τ_crit=4.7, frame, a^{-1}, no-go, domain

§13. Conclusion

ZS-M12 establishes Auto-Surgery and traces its consequences to the inflation initial condition problem. Six principal results:

(1) Continuous flow $d\Phi/d\tau = i^N - \Phi$, z* unique attractor (DERIVED + PROVEN).
(2) σ-HSI Locking Theorem: z* = conformal ∩ stroboscopic (DERIVED-CONDITIONAL).
(3) Topological cap $\Omega^2 = 1.0258$: singularity resolution (DERIVED).
(4) No-Go: minimal model cannot launch inflation (PROVEN + DERIVED).
(5) Centrifugal barrier: $\varepsilon_{min} = (Q^2/\lambda)^{1/6} \approx 30.7$, weakens as a^{-1} (DERIVED-CONDITIONAL, contingent on V1).
(6) $T_c \approx T_{reh} \approx 2.5\times10^{15}$ GeV: cyclic self-consistency (DERIVED).

The complete causal chain:

$$\mathbf{A = 35/437 \rightarrow \delta\varphi = A \rightarrow Q = A \rightarrow \varepsilon_{min} = (Q^2/\lambda)^{1/6} \approx 30.7 \gg 2.64} \qquad \mathbf{(6)}$$

Zero new parameters. If V1 (Planck-bulk matching) is confirmed, the inflation initial condition problem—shared by all slow-roll models—is resolved within Z-Spin by the same constant that governs the Hubble tension, baryon asymmetry, and gauge couplings.

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Code Availability

verify_zs_m12.py (GitHub: KennyKang-git/zspin/verify_scripts/)

Appendix A: Detailed Derivations

A.1 Flow Equation Chain

The HSI Theorem (ZS-M1) derives $T(z) = i^z$ from (a) conformal factor $\Omega^2 = 1+A\varepsilon^2$ (ZS-F1), (b) $\hat{W}^2=I$ involution (ZS-F5), and (c) Q=11=(2,3,6) sector decomposition. Weyl additivity on $(\mathbb{C},+)$ and multiplicativity on $(\mathbb{C}^*,\times)$ force $T(z) = \exp(\alpha z)$; the quarter-turn condition fixes $\alpha = i\pi/2$. Forward-Euler with $h = \tau P$ gives $F(\Phi) = (i^N - \Phi)/\tau P$, Eq. (1) in Planck units.

A.2 Stability Analysis

$J = (i\pi/2)\cdot z^* - 1 = -1.5664 + 0.6885i$. $|\mathrm{Re}(\lambda)| = 1.5664$, $|\mathrm{Im}(\lambda)| = 0.6885$. Quality factor $Q = 0.220$ (overdamped). $\tau_{1/2} = \ln 2/1.5664 = 0.44$, $\tau_{\square 9} = \ln 100/1.5664 = 2.94$.

A.3 Centrifugal Barrier Derivation (Corrected 1/6 Exponent)

$V'\mathrm{eff} = \lambda(\varepsilon^2-1)\varepsilon - Q^2/(a^6\varepsilon^3) = 0$. Multiply by ε^3: $\lambda\varepsilon^4(\varepsilon^2-1) = Q^2/a^6$, giving $\lambda\varepsilon^6 - \lambda\varepsilon^4 = Q^2/a^6$. For $\varepsilon \gg 1$: $\lambda\varepsilon^6 \approx Q^2/a^6$, hence $\varepsilon\mathrm{min} = (Q^2/(\lambda a^6))^{1/6} = (Q^2/\lambda)^{1/6}/a$. The exponent is 1/6 (not 1/4, which arises from the incorrect small-ε approximation). Numerical verification: root of $V'\mathrm{eff} = 0$ at $\varepsilon = 30.73$ vs analytical 30.72 (0.02% agreement).

A.4 Threshold and Robustness

From $\varepsilon_{sr}^6 = Q^2_{th}/\lambda$: $Q_{th} = \sqrt{\lambda} \times \varepsilon^3_{sr} = 5.08\times10^{-5}$. Margin: $A/Q_{th} = 1{,}576$. Dissipation tolerance: $1-1/1576 = 99.94\%$. Critical time: $\tau_{crit} = \ln(1576)/1.566 = 4.7\ \tau_P$. Surgery window: 1–3 τ_P. Margin: 1.6×.

A.5 H6 Rejection: Energy Decomposition

For $\Phi_0 = 1$: $\int T_{radial} d\tau = 0.081$ (37.4%), $\int T_{angular} d\tau = 0.136$ (62.6%). Both decay to zero. Angular→radial transfer: 3.9% of timesteps, reverse 4.0%. Net: -9×10^{-5}. Total KE at $\tau\to\infty$: 0 (machine precision). $\mathrm{Re}(\lambda) < 0$ guarantees monotonic envelope decay. [DISPROVEN]

A.6 Research Narrative

The centrifugal barrier emerged from a chain of investigations. The Perelman analogy motivated the flow equation and σ-HSI theorem. The No-Go proved the minimal model fails. Five hypotheses (H2, H5, H6, H7, H2+H5) were systematically rejected through quantitative computation. The Quantum Foam Engine revealed symmetry restoration and $T_c \approx T_{reh}$. Domain wall analysis then uncovered the centrifugal barrier: U(1) angular momentum from Z-Telomere creates $\varepsilon_{min} = (Q^2/\lambda)^{1/6} \approx 30.7$, resolving inflation IC contingent on V1. This Perelman → quantum foam → centrifugal barrier trajectory exemplifies how systematic falsification leads to unexpected solutions.

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Version History

v1.0 (March 2026): Initial public release. Flow equation, σ-HSI Locking Theorem, Auto-Surgery, No-Go Theorem, Quantum Foam Engine, centrifugal barrier ($\varepsilon_{min} = (Q^2/\lambda)^{1/6} \approx 30.7$, launch-then-release), L robustness (1,576× margin, $\tau_{crit} = 4.7\ \tau_P$), domain wall resolution, $T_c \approx T_{reh}$ coincidence, five rejected hypotheses, V1 OPEN. 35/35 PASS. (Consolidated from internal Z-Spin Collaboration research notes, March 2026)